

Lecture 6

Microeconomic Theory 3 – ECON 709

Fall 2026

Recap' lecture 5

- ▶ We proved Debreu's result:

Assume X is a convex subset of $\mathbb{R}^n$. Let $\succsim$ be a continuous and rational preference relation on X . Then there is a continuous utility function $u(x)$ which represents $\succsim$.

With the simplifying assumptions of monotonicity and $X = \mathbb{R}_+^K \dots$

- ▶ If $\succsim$ on X is represented by a continuous utility function u , then $\succsim$ is rational and continuous.

Having talked about tastes, attitudes, and opinions, we now turn to behavior...

Choice

based on: Rubinstein, lecture 3; MWG, sections 1.C and 1.D

Choice Problem and Context

- ▶ X is the grand set of possible alternatives
- ▶ A is a non-empty subset of X , which constitutes the *choice problem*: Which element of A is chosen?
- ▶ D is the collection of all possible choice problems $A \subseteq X$ which the decision maker could face, thus D is a set of nonempty subsets of X . D does not need to include all possible nonempty subsets of X , but only the relevant choice problems
- ▶ we will refer to a pair (X, D) as a *context*

Choice Problem and Context: Example

- ▶ suppose we are interested in a student's choice from the set of universities to which he has been admitted
- ▶ let $X = \{x_1, \dots, x_N\}$ be the set of all universities to which the student applied to
- ▶ a choice problem $A \subseteq X$ is interpreted as the set of universities to which he has been admitted
- ▶ if the fact that the student was admitted to some universities does not mean he will be admitted to others, then D contains the $2^N - 1$ nonempty subsets of X
- ▶ if, for example, the universities are listed according to difficulty of getting into (x_1 being the most difficult) and if the fact that the student is admitted to x_k means that he is admitted to all less "prestigious" universities, that is, to all x_l with $l > k$, then D consists of the N sets $A_1, \dots, A_N$ where $A_k = \{x_k, \dots, x_N\}$.

Choice Rule

- ▶ given a context (X, D) , a choice rule is a correspondence C which assigns to every $A \in D$ a nonempty subset of chosen elements $C(A) \subseteq A$, that is $\emptyset \neq C(A) \subseteq A$
- ▶ $C(A)$ is the non-empty set of acceptable choices for the decisionmaker when he is presented with choice problem A
 - ▶ If $C(A)$ is a singleton, then this single element is the DM's choice. Back to our example: If $C(\{McGill, Toronto, Vancouver\}) = \{McGill\}$, then McGill is your choice when admitted to $\{McGill, Toronto, Vancouver\}$.
 - ▶ If $C(A)$ contains more than one element, then this does not mean that the DM chooses all of the elements in $C(A)$ at once, but rather each element is an equally acceptable good choice for the DM when presented with A . No unique choice prediction. "Equilibrium" interpretation: If some $x \in C(A)$ is chosen, then the decision maker has no desire to move away from his choice; he would, however, always swap an $x \notin C(A)$ for an $x \in C(A)$. Back to our example: If $C(\{McGill, Toronto, Vancouver\}) = \{McGill, Toronto\}$, then Toronto and McGill are equally acceptable choices for you.

Rational Choice

- ▶ In economics we usually think of an agent's decision process as a *rational choice* which goes through the following stages:
 1. the agent has a rational preference relation $\succsim$ over the set of all possible alternatives X
 2. the set of possible choice problems D is determined
 3. a choice problem $A \in D$ realizes
 4. the agent's choice is from the set of her most desirable alternative(s) in A , that is, one of the maximal elements in A given $\succsim$, a $\succsim$ -best/optimal choice from A
- ▶ Note that the
 - ▶ Rational economic agent's preferences are shaped before D and A are recognized. $\succsim$ over X is taken as independent of D and A .
 - ▶ Rationality in economics does not involve making any judgments about preferences. A rational agent can have preferences that are universally viewed as being against his best interest.
- ▶ Idea of rational choice reflected in the following definitions...

Choice rule derived from a given preference relation

Definition 6

Given a preference relation $\succsim$, the *induced choice rule* $C_{\succsim}$ is defined by

$$C_{\succsim}(A) = \{x \in A \mid x \succsim y \text{ for all } y \in A\}.$$

- ▶ given $C_{\succsim}$, the decision maker will choose an object that he weakly prefers to all other objects in A
- ▶ in principle, we could have $C_{\succsim}(A) = \emptyset$ for some A
- ▶ however, if $\succsim$ is rational and X is finite, then $C_{\succsim}(A)$ is never empty
- ▶ this follows directly from the fact that if $\succsim$ is rational, then every finite $A \subseteq X$ has a maximal element (see Lemma 1), and a maximal element is in $C_{\succsim}(A)$

Rationalizability of a Choice Rule

Definition 7

A choice rule C is **rationalizable** if there is a rational preference relation $\succsim$ on X such that $C = C_{\succsim}$ (i.e., $C(A) = C_{\succsim}(A)$ for any A in the domain of C).

A preference relation $\succsim$ **rationalizes** C if $C = C_{\succsim}$ and $\succsim$ is rational (i.e., $C(A) = C_{\succsim}(A)$ for any A in the domain of C).

- ▶ a choice rule is rationalizable if she maximizes some rational preference relation among the available alternatives
- ▶ there may be more than one preference relation which rationalizes a given choice rule

Preferences derived from choices

Definition 8

Given a choice rule C , its *revealed preference relation* $\succsim_C$ is defined by

$x \succsim_C y$ if and only if $x, y \in A$ and $x \in C(A)$ for some $A \in D$.

- ▶ we read $x \succsim_C y$ as "x is revealed at least as good as y", if in addition $y \notin C(A)$, we say that "x is revealed strictly better than y"
- ▶ x is revealed at least as good as y iff x is an acceptable choice when y is available, and if in addition, y is not an acceptable choice, then x is revealed strictly better than y
- ▶ note that $\succsim_C$ is not necessarily rational, in particular for $\succsim_C$ to be complete, we must have that for any pair $x, y \in X$,
 $x, y \in A$ and $x \in C(A)$ or $y \in C(A)$ for some $A \in D$

A condition for C to be rationalizable: The Weak Axiom

Definition 9

The choice rule C satisfies the *weak axiom of revealed preference* if the following property holds:

If $x, y \in A \cap B$ and $x \in C(A)$ and $y \in C(B)$, then $x \in C(B)$.

- ▶ In words: if x is an acceptable choice when y is available, then there can be no A containing both x and y for which y is an acceptable choice and x is not
- ▶ An alternative formulation of the weak axiom through the revealed preference relation:
If x is revealed at least as good as y in choice problem A , then y cannot be revealed strictly better than x in choice problem B .

The Weak Axiom implies Sen's α

Definition 10 (Sen's α .)

*A choice rule C satisfies Sen's α if the following property holds:
If $x \in A \subseteq B$ and $x \in C(B)$, then $x \in C(A)$.*

- ▶ "Sen's α " is also referred to as "Independence of Irrelevant Alternatives".
- ▶ Example: If I choose Italy out of all worldwide holiday spots ($I \in C(\text{World})$), then I also choose Italy out of all European holiday spots ($I \in C(\text{Europe})$).
- ▶ If a choice rule C satisfies the WA, then it satisfies Sen's α :
If $x \in C(B)$, then x is revealed at least as good as every $y \in A \subseteq B$. Thus, by the WA, no $y \in A$ can be revealed strictly better than x and since $C(A)$ is non-empty, x must be in $C(A)$.

The Weak Axiom implies Sen's β

Definition 11 (Sen's β .)

A choice rule C satisfies Sen's β if the following property holds:

If $x, y \in A \subseteq B$, $x, y \in C(A)$ and $y \in C(B)$, then $x \in C(B)$.

- ▶ "Sen's β " is sometimes referred to as "Expansion Consistency".
- ▶ Example: If France and Italy are the best European holiday spots for me ($F, I \in C(Europe)$) and Italy is one of the best worldwide holiday spots for me ($I \in C(World)$), then France must also be a best worldwide holiday spot for me ($F \in C(World)$).
- ▶ If a choice rule C satisfies the WA, then it satisfies Sen's β :
Since $x, y \in C(A)$, x is revealed at least as good as y . By the WA, y cannot be revealed strictly better than x . Thus x must be in $C(B)$ as well.

A choice rule induced from a rational preference relation satisfies the WA.

Proposition 10

Suppose that $\succsim$ is a rational preference relation. Then the induced choice rule $C_{\succsim}$ satisfies the weak axiom.

Proof.

Suppose we have $x, y \in A \cap B$ and $x \in C_{\succsim}(A)$ and $y \in C_{\succsim}(B)$. By the definition of $C_{\succsim}$, this implies that $x \succsim y$ and $y \succsim x$, and $y \succsim z$ for all $z \in B$. By transitivity, we know that $x \succsim z$ for all $z \in B$. Then $x \in C_{\succsim}(B)$ by the definition of $C_{\succsim}$. Thus $C_{\succsim}$ satisfies the WA. □

- ▶ Remember now that a choice rule C is rationalizable if there is a rational preference relation $\succsim$ such that $C_{\succsim} = C$ on the domain of C .
- ▶ Thus, Prop. 10 implies that a choice rule must satisfy the weak axiom in order to be rationalizable.
- ▶ The weak axiom is a necessary condition for rationalizability.

WA is sufficient for rationalizability only with additional assumptions on D .

Example:

- ▶ Suppose that $X = \{x, y, z\}$, $D = \{\{x, y\}, \{y, z\}, \{x, z\}\}$ and $C(\{x, y\}) = \{x\}$, $C(\{y, z\}) = \{y\}$ and $C(\{x, z\}) = \{z\}$.
- ▶ C trivially satisfies WA because there are no distinct $x, y \in X$ and $A, B \in D$ such that $x, y \in A \cap B$.
- ▶ There is no preference relation which rationalizes C :
 $C(\{x, y\}) = \{x\}$ and $C(\{y, z\}) = \{y\}$ imply that it would be necessary to have $x \succ y$ and $y \succ z$. Transitivity of $\succsim$ would then require that $x \succ z$ but this contradicts $C(\{x, z\}) = \{z\}$.

Intuition:

- ▶ The more subsets of X are contained in D , the more WA restricts a choice rule C
and if D includes enough subsets of X , then the restrictions of WA on C are strong enough for C to be rationalizable.
- ▶ With too few subsets of X in D , WA might not rule out choice behavior which is incompatible with a rational preference relation.
- ▶ E.g. in above example, including X in D would make a choice rule which satisfies WA rationalizable.

WA as a sufficient condition for rationalizability

Proposition 11

Assume that C is a choice rule which satisfies the weak axiom and that D includes at least all subsets of X of size at most 3. Then C is rationalizable, that is, there exists a rational preference relation $\succsim$ such that $C = C_{\succsim}$ on the domain of C . Moreover, $\succsim$ is the unique preference relation which rationalizes C .

Proof.

A natural candidate for a rationalizing preference relation is the revealed preference relation $\succsim_C$. We show first that $\succsim_C$ is rational, second that $C = C_{\succsim_C}$ on the domain of C , and third that no other preference relation rationalizes C .

1. Reflexivity and completeness: Since C is non-empty, for all $x, y \in X$, $x \in C(\{x, y\})$ or $y \in C(\{x, y\})$. Thus $x \succsim_C x$ for all $x \in X$, and $x \succsim_C y$ or $y \succsim_C x$ for any distinct $x, y \in X$.

Transitivity: Let $x \succsim_C y$ and $y \succsim_C z$. Since $C(\{x, y, z\}) \neq \emptyset$, x, y , or z in $C(\{x, y, z\})$. If $y \in C(\{x, y, z\})$, then $x \in C(\{x, y, z\})$ by the WA. If $z \in C(\{x, y, z\})$, then $y \in C(\{x, y, z\})$ by the WA and hence $x \in C(\{x, y, z\})$. Thus in any case $x \in C(\{x, y, z\})$ and $x \succsim_C z$ by the definition of $\succsim_C$.

2. Suppose $x \in C(B)$. Then $x \succsim_C y$ for all $y \in B$ and thus $x \in C_{\succsim_C}(B)$.

Suppose $x \in C_{\succsim_C}(B)$. This implies $x \succsim_C y$ for all $y \in B$. Since $C(B) \neq \emptyset$, there is $y \in C(B)$. If $y \neq x$, then $x \in C(B)$ because y cannot be revealed strictly better than x by WA. Thus, $x \in C_{\succsim_C}(B)$ if and only if $x \in C(B)$, that is, $C = C_{\succsim_C}$.

3. D includes all two-element subsets of X . Hence the choice behavior in C completely determines the pairwise preference relations over X of a rationalizing preference relation; no other than the revealed preference relation can be rationalizing. $\square$

So far we have discussed preferences and choice in an abstract setting. Now, we are going to move to a special case of an economic agent in more detail: preferences and choice of a consumer.

Consumer Preferences

based on: Rubinstein, lecture 4; MWG, sections 2.A-2.C, 3.A-3.C

The consumer's world

- ▶ the consumer lives in a market economy with commodities (goods and services) that can be obtained in exchange for money at known prices (or equivalently, in exchange for other commodities at known exchange rates)
- ▶ the consumer makes choices between different available combinations of commodities
- ▶ K commodities
- ▶ a *bundle* $x = (x_1, \dots, x_K)$ is a combination of the K commodities where x_k is the quantity of commodity k
- ▶ the set of all available alternatives, X , is the *consumption set*: the set of all bundles which can potentially be consumed given the physical constraints of the environment
- ▶ assume that $X = \mathbb{R}_+^K = \{x = (x_1, \dots, x_K) \mid \text{for all } k, x_k \geq 0\}$ (all non-negative bundles of commodities), unless explicitly stated otherwise

Consumer's preferences over X

- ▶ from now on, we assume that a consumer's preference relation $\succsim$ over X is rational
- ▶ we will discuss additional properties of $\succsim$ in the following, which rely on the particular nature of X :
 - ▶ monotonicity: relies on the ability to compare bundles according to the amount of any particular commodity
 - ▶ convexity: relies on the ability to speak of the sum of two bundles and the multiplication of a bundle by a scalar
- ▶ we will look at special classes of preference relations: homothetic, quasi-linear
- ▶ we will often demonstrate properties of the consumer's preferences by referring to the map of indifference curves, where an indifference curve is a set of the type $\{y | y \sim x\}$ for some bundle x

Monotonicity: "more is better"

Definition 12

The preference relation $\succsim$ on X is *monotone* if for all $x, y \in X$, $x_k \geq y_k$ for all k implies $x \succsim y$, and $x_k > y_k$ for all k implies $x \succ y$.

The preference relation $\succsim$ on X is *strongly monotone* if for all $x, y \in X$, $x_k \geq y_k$ for all k and $x \neq y$ implies $x \succ y$.

- Strong monotonicity obviously implies monotonicity.

- ▶ Monotonicity allows for indifference between bundles in which some quantities are equal. E.g. if $x_1 > y_1$ and $x_2 = y_2$, then $x \succsim y$ and $y \succsim x$ as well is not ruled out. Thus we can have $(x_1, x_2) \sim (y_1, y_2)$. This would not be possible with strong monotonicity.
- ▶ Our definition of monotonicity is a stronger version of MWG, who do not include the requirement that if $x_k \geq y_k$ for all k , then $x \succsim y$. But of course if preferences are monotone according to our definition, then they are monotone according to MWG.

Monotonicity and Utility Representations

- ▶ A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is strictly increasing if $x_k \geq y_k$ for all k implies $f(x) \geq f(y)$, and if $x_k > y_k$ for all k implies $f(x) > f(y)$.

A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is strongly increasing if $x_k \geq y_k$ for all k and $x \neq y$ implies $f(x) > f(y)$.

The following result is easy to check:

- ▶ If u represents preference relation $\succsim$,
 - ▶ $\succsim$ is monotone if and only if u is strictly increasing.
 - ▶ $\succsim$ is strongly monotone if and only if u is strongly increasing.

Examples of (strongly) monotone preference relations

- ▶ If a preference relation $\succsim$ on $X = \mathbb{R}_+^2$ is represented by $u(x) = \min \{x_1, x_2\}$, then $\succsim$ is monotone, but not strongly monotonic.
- ▶ If a preference relation $\succsim$ on $X = \mathbb{R}_+^2$ is represented by $u(x) = x_1 + x_2$, then $\succsim$ strongly monotone.

Strong monotonicity implies strictly decreasing indifference curves

Let $K = 2$. If $y \sim z$ and $y_1 < z_1$, then $z_2 < y_2$ by strong monotonicity. If $z_2 \geq y_2$, then strong monotonicity would imply that $z \succ y$, a contradiction.

